



## Solve with us 2.2 - Not Graded



This assignment will not be graded and is only for practice.

**Key Points: 1**

**A matrix is in row echelon form if :**

- The first non-zero element (the leading entry) in a row is 1.
- The column containing the leading 1 of a row is to the right of the column containing the leading 1 of the row above it.
- Any non-zero rows are always above rows with all zeros.

**A matrix is in reduced row echelon form if :**

- The first non-zero element in the first row (the leading entry) is the number 1.
- All subsequent non-zero rows must also have their leading entries (i.e. first non-zero entries) as 1 and they should appear to the right of the leading entry in the previous row.
- The leading entry in each row must be the only non-zero number in its column.
- Any non-zero rows are always above rows with all zeros.

**Note: Any matrix which is in reduced row echelon form is also in row echelon form.**

1 point

Choose the correct options for the matrix  $A = \begin{bmatrix} 1 & 2 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}$   $A = \begin{bmatrix} & & \\ 1 & 0 & 2 & 0 & 0 & 1 & 0 \\ & & & & & & \end{bmatrix}$

[Hint: Find out the leading entries for each rows and observe whether the matrix is satisfying the conditions of row echelon form and reduced row echelon form.]

☐

AA is in row echelon form but not in reduced row echelon form.

☐

AA is in both row echelon form and reduced row echelon form.

☐

AA is neither in row echelon form nor reduced row echelon form.

**No, the answer is incorrect.**

**Score: 0**

**Accepted Answers:**

AA is in both row echelon form and reduced row echelon form.

1 point

Choose the correct options for the matrix  $A = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}$   $A = \begin{bmatrix} & & \\ 0 & 1 & 0 & 1 & 0 & 1 & 0 \\ & & & & & & \end{bmatrix}$

☐

AA is in row echelon form but not in reduced row echelon form.

☐

AA is in both row echelon form and reduced row echelon form.

☐

AA is neither in row echelon form nor reduced row echelon form.



**No, the answer is incorrect.**

**Score: 0**

**Accepted Answers:**

AA is neither in row echelon form nor reduced row echelon form.

1 point

Choose the correct options for the matrix  $F = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

☐

AA is in row echelon form but not in reduced row echelon form.

☐

AA is in both row echelon form and reduced row echelon form.

☐

AA is neither in row echelon form nor reduced row echelon form.

**No, the answer is incorrect.**

**Score: 0**

**Accepted Answers:**

AA is in row echelon form but not in reduced row echelon form.

**Key Points: 2**

Let  $Ax = b$  be a system of linear equations and suppose  $AA$  is in reduced row echelon form. Assume that for every zero row of  $A$ , the corresponding entry of  $b$  is also 0 (i.e., if the  $i$ -th row of  $AA$  is 00, then so is  $b_i$ .)

- If the  $i$ -th column has the leading entry of some row, we call  $x_i$  a dependent variable.
- If the  $i$ -th column does not have the leading entry of any row, we call  $x_i$  an independent variable.

1 point

Consider a system of linear equations:

$$\begin{aligned} 0x_1 + x_2 + 0x_3 + 0x_4 &= 1 \\ 0x_1 + 0x_2 + x_3 + 0x_4 &= 1 \end{aligned} \quad \begin{aligned} 0x_1 + x_2 + 0x_3 + 0x_4 &= 10 \\ 0x_1 + 0x_2 + x_3 + 0x_4 &= 1 \end{aligned}$$

Choose the set of correct options.

[Hint: Recall the definitions of independent and dependent variable with respect to reduced row echelon form.]

☐

$x_1$  and  $x_2$  are dependent variables.

☐

$x_2$  is a dependent variable.

☐

$x_3$  and  $x_4$  are independent variables.

☐

$x_4$  is an independent variable.

**No, the answer is incorrect.**

**Score: 0**

**Accepted Answers:**

$x_2$  is a dependent variable.

$x_4$  is an independent variable.

1 point

let  $Ax = b$  be a system of linear equations, where  $A = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$ , and

$$b = \begin{bmatrix} 3 \\ 2 \\ 0 \\ 0 \end{bmatrix}$$

Choose the set of correct options.

☐

$x_1$  is a dependent variable.

☐

$x_2$  is a dependent variable.

☐

$x_3$  is a dependent variable.

☐

The expressions  $x_1 = 3$  and  $x_3 = 2 - x_4$  show the dependency of  $x_1$  and  $x_3$  on the elements of  $b$  and the independent variables.

**No, the answer is incorrect.**

**Score: 0**

**Accepted Answers:**

$x_1$  is a dependent variable.

$x_3$  is a dependent variable.

The expressions  $x_1 = 3$  and  $x_3 = 2 - x_4$  show the dependency of  $x_1$  and  $x_3$  on the elements of  $b$  and the independent variables.

**Key Points: 3**

The three different types of elementary row operations that can be performed on a matrix are:

- **Type 1:** Interchanging two rows.
- **Type 2:** Multiplying a row with some constant.
- **Type 3:** Adding a scalar multiple of a row to another row.

Using these row operations one can obtain row echelon form or reduced row echelon form of a given matrix.

Consider the following matrix A and answer the questions (a), (b), (c), (d), and (e).

$$A = \begin{bmatrix} 1 & 3 & 2 & 0 \\ 0 & 2 & 1 & 0 \\ 2 & 4 & 4 & 3 \end{bmatrix}$$

1 point

What will be the third row of the matrix if we perform the row operation  $R_3 - 2R_1$  on A?

☐

(1123)

☐

(41083)(41083)

☐

(0-203)(0-203)

☐

(0443)(0443)

**No, the answer is incorrect.****Score: 0****Accepted Answers:**

(0-203)(0-203)

*1 point*What will be the second row of the matrix if we perform the row operation  $\frac{1}{2}R_2 \rightarrow 2R_2$  on  $AA$ ?☐

(0110)(0110)

☐

(0420)(0420)

☐(01 $\frac{1}{2}$ 0)(01210)☐(122 $\frac{3}{2}$ )(12223)**No, the answer is incorrect.****Score: 0****Accepted Answers:**(01 $\frac{1}{2}$ 0)(01210)*1 point*If we perform the row operations mentioned in question (a) and question (b), simultaneously on  $AA$ , the matrix (let it be called  $BB$ ) we get is:☐

$$\begin{bmatrix} 1 & 3 & 20 \\ 0 & 1 & \frac{1}{2}0 \\ 0 & -203 \end{bmatrix} \begin{bmatrix} 10031-22210003 \end{bmatrix}$$
☐

$$\begin{bmatrix} 1 & 3 & 20 \\ 0 & 1 & \frac{1}{2}0 \\ 1 & 1 & 23 \end{bmatrix} \begin{bmatrix} 101311212003 \end{bmatrix}$$
☐

$$\begin{bmatrix} 1 & 3 & 20 \\ 0 & 4 & 20 \\ 0 & -203 \end{bmatrix} \begin{bmatrix} 10034-2220003 \end{bmatrix}$$
☐

$$\begin{bmatrix} 1 & 3 & 20 \\ 0 & 1 & \frac{1}{2}0 \\ 0 & 4 & 43 \end{bmatrix} \begin{bmatrix} 1003142214003 \end{bmatrix}$$

**No, the answer is incorrect.**

**Score: 0**

**Accepted Answers:**

$$\begin{array}{c} 1 \ 3 \ 20 \\ [0 \ 1 \ 7] \\ 0 \ -203 \end{array} \left[ \begin{array}{c} \\ \\ 10031-22210003 \end{array} \right]$$

1 point

Which row operations have to be performed on the matrix  $BB$  (matrix  $BB$  is as in question (c)) to make all the entries 0 in the second column, except the leading entry of the second row.

☐

$R_1 - 3R_2$   $R_1 - 3R_2$  and  $R_3 - 2R_2$   $R_3 - 2R_2$

☐

$R_1 + 3R_2$   $R_1 + 3R_2$  and  $R_3 + 2R_2$   $R_3 + 2R_2$

☐

$R_1 + 3R_2$   $R_1 + 3R_2$  and  $R_3 - 2R_2$   $R_3 - 2R_2$

☐

$R_1 - 3R_2$   $R_1 - 3R_2$  and  $R_3 + 2R_2$   $R_3 + 2R_2$

**No, the answer is incorrect.**

**Score: 0**

**Accepted Answers:**

$R_1 - 3R_2$   $R_1 - 3R_2$  and  $R_3 + 2R_2$   $R_3 + 2R_2$

1 point

The reduced row echelon form of  $AA$  is

[Hint: Recall the definition of reduced row echelon form.]

☐

$$\begin{array}{c} 10 \ 10 \\ [0 \ 1 \ 7] \\ 00 \ 13 \end{array} \left[ \begin{array}{c} \\ \\ 10001021211003 \end{array} \right]$$

☐

$$\begin{array}{c} 1000 \\ [0 \ 1 \ 7] \\ 00 \ 13 \end{array} \left[ \begin{array}{c} \\ \\ 100010001003 \end{array} \right]$$

☐

$$\begin{array}{c} 100 \ 0 \\ [0 \ 10 \ -3] \\ 00 \ 1 \ 3 \end{array} \left[ \begin{array}{c} \\ \\ 1000100010-233 \end{array} \right]$$

☐

$$\begin{array}{c} 100 \ -3 \\ [0 \ 10 \ -3] \\ 00 \ 1 \ 3 \end{array} \left[ \begin{array}{c} \\ \\ 100010001-23-233 \end{array} \right]$$

**No, the answer is incorrect.**

**Score: 0**

**Accepted Answers:**

$$\begin{bmatrix} 100 & -3 \\ 010 & -3 \\ 001 & 3 \end{bmatrix} \begin{bmatrix} 100010001 & -23 & -233 \end{bmatrix}$$

**Key Points: 4**

Let  $Ax = b$  denote the matrix representation of a system of linear equations.

- The augmented matrix is denoted by  $[A | b]$ .
- The matrix obtained by performing the operations that transform  $A$  to its reduced row echelon form  $R$  on the augmented matrix is denoted by  $[R | c]$ .
- The solutions of  $Rx = c$  are the same as the solutions of  $Ax = b$ .

Consider a system of linear equations

$$\begin{aligned} 2x_1 + x_2 &= 1 - x_1 + x_3 + x_4 = -1x_1 + x_2 - x_3 + x_4 = 2 - x_1 + x_3 + x_4 = 1. \\ 2x_1 + x_2 &= 1 - x_1 + x_3 + x_4 = -1x_1 + x_2 - x_3 + x_4 = 2 - x_1 + x_3 + x_4 = 1. \end{aligned}$$

Answer the questions (a), (b) and (c) based on the above information.

1 point

Which of the following represents an augmented matrix of the system?

[Hint: First try to write down the coefficient matrix  $A$  and the corresponding vector  $b$ .]

- ☐
- ☐
- ☐
- ☐

**No, the answer is incorrect.**

**Score: 0**

**Accepted Answers:**

1 point

What will be the matrix  $[R | c]$  obtained by performing the operations that transform  $A$  to its reduced row echelon form  $R$  on the augmented matrix?

- ☐
- ☐
- ☐
- ☐

**No, the answer is incorrect.**

**Score: 0**

**Accepted Answers:**

1 point

Choose the correct option. [Hint: Write down the system of linear equations corresponding to the reduced row echelon form of the augmented matrix.]

- ☐ The system of linear equations has an infinite number of solutions.
- ☐ The system of linear equations has a unique solution.
- ☐ The system of linear equations has no solution.
- ☐

$x_1 = x_2 = x_3 = x_4 = 1$  is a solution of the system of linear equations.

**No, the answer is incorrect.**

**Score: 0**

**Accepted Answers:**

The system of linear equations has no solution.

[Check Answers](#)